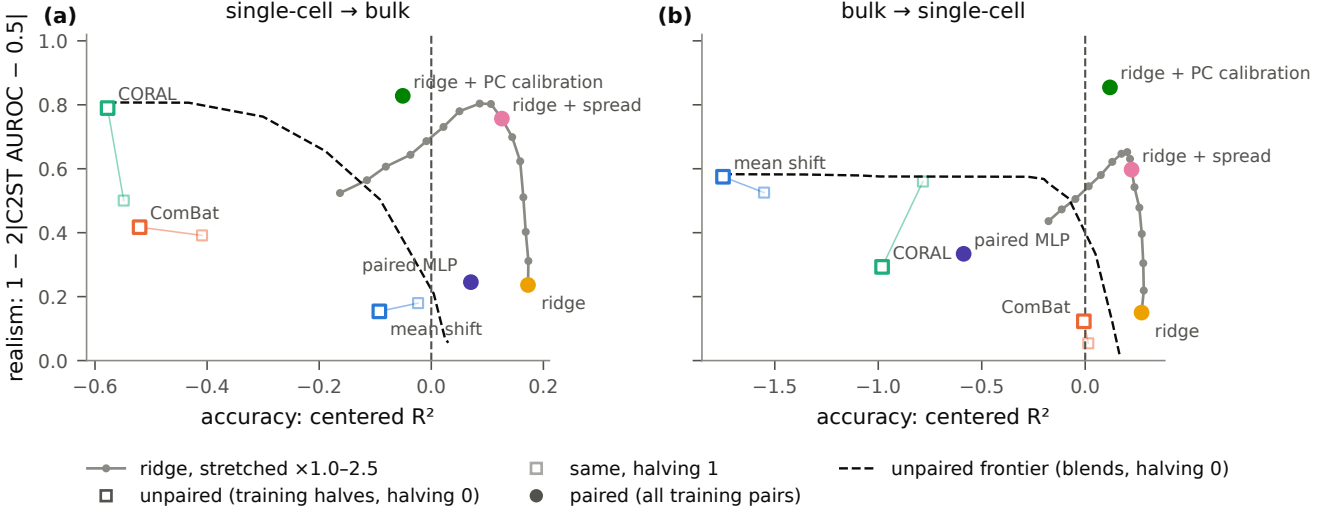
(a)
single-cell → bulk
realism: 1 − 2|C2ST AUROC − 0.5|
accuracy: centered R²
CORAL
ComBat
ridge + PC calibration
ridge + spread
paired MLP
ridge
mean shift
(b)
bulk → single-cell
accuracy: centered R²
mean shift
ridge + PC calibration
ridge + spread
CORAL
paired MLP
ComBat
ridge
ridge, stretched ×1.0–2.5
unpaired (training halves, halving 0)
same, halving 1
paired (all training pairs)
unpaired frontier (blends, halving 0)